

Global Counterspace Capabilities: An Open Source Assessment

Victoria Samson, Secure World Foundation

Quad Nations Security Dialogue, Sydney, Australia

Sept. 28, 2025



SECURE
WORLD
FOUNDATION

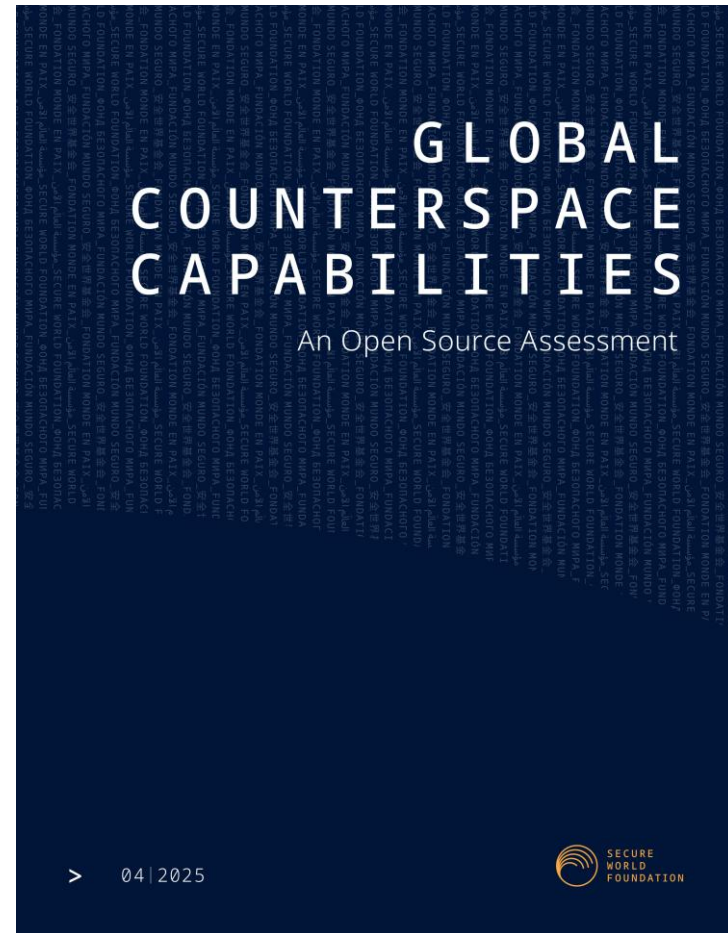


About Secure World Foundation

- **SWF** is a private operating foundation that promotes cooperative solutions for space sustainability
- **Vision:** The secure, sustainable, and peaceful uses of outer space that contribute to global stability on Earth
- **Our mission:** Work with governments, industry, international organizations, and civil society to develop and promote ideas and actions to achieve the secure, sustainable, and peaceful uses of outer space benefiting Earth and all its peoples



- Existence of counterspace capabilities is not new, but the circumstances surrounding them are
- Significant R&D/testing of a wide range of destructive & non-destructive counterspace capabilities by multiple countries
- ***Only non-destructive capabilities are actively being used in current military operations***



<https://swfound.org/counterspace>



Counterspace Capabilities

Co-orbital: systems that are placed into orbit and then maneuver to approach the target to attack it by various means, including destructive and non-destructive

Direct Ascent: systems that use ground, air-, or sea-launched missiles with interceptors that are used to kinetically destroy satellites through force of impact, but are not placed into orbit themselves

Directed Energy: systems that use focused energy, such as laser, particle, or microwave beams to interfere or destroy space systems

Electronic Warfare: systems that use radiofrequency energy to interfere with or jam the communications to or from satellites

Cyber: systems that use software and network techniques to compromise, control, interfere, or destroy computer systems

Space Situational Awareness: knowledge about the space environment and human space activities that enables both offensive and defense counterspace operations



2025 Global Assessment

	US	RUSSIA	CHINA	INDIA	AUS.	FRANCE	IRAN	ISRAEL	JAPAN	N. KOREA	S. KOREA	UK
LEO Co-Orbital	■	▲	▲	●	●	■	●	●	●	●	●	●
MEO/GEO Co-Orbital	■	■	■	●	●	■	●	●	●	●	●	●
LEO Direct Ascent	■	▲	▲	■	●	●	●	●	●	●	●	●
MEO/GEO Direct Ascent	■	■	■	●	●	●	●	●	●	●	●	●
Directed Energy	■	■	■	●	●	■	●	●	●	●	●	●
Electronic Warfare	▲	▲	▲	■	■	■	■	▲	■	■	●	●
Space Situational Awareness	▲	▲	▲	■	■	■	■	■	■	■	■	■

LEGEND: NONE ● SOME ■ SIGNIFICANT ▲ UNCERTAIN ? NO DATA —



Case Study: Russia's Reported Nuclear ASAT

- In spring 2024, news emerged that the United States believes Russia to be working on a nuclear warhead that could be placed into space and used as an ASAT
 - To be clear: there is a satellite on orbit thought to be associated with this program – Cosmos 2553 – but no nuclear warhead is in orbit
- A nuclear detonation in space would have two types of effects
 - First is exposure to X-ray, gamma, and ultraviolet photos from the detonation, occurring instantly and severely damaging or completely disabling any satellite within line-of-sight
 - Second and more enduring is radiation exposure from trapped ionized particles in the Earth's magnetic field & amplified radiation in the Van Allen belts; as satellites pass through these trapped particles, they would eventually stop working unless they have been hardened against nuclear attacks
 - We know this because both the US and USSR tested nuclear warheads in space in the early years of the Cold War
- This is against the Outer Space Treaty's article 4, which prohibits placing weapons of mass destruction in orbit; also relevant are the 1963 LTBT and 1978 Environmental Modification Convention



Case Study: Military RPO Activities

- United States, Russia, and China have all conducted multiple tests of satellite technologies for robotic rendezvous and proximity operations (RPO)
 - Done largely to support surveillance and inspection of other space objects in both LEO and GEO, most of which are related to military or intelligence operations
 - US' X-37B and China's Shenlong spaceplanes have both conducted RPOs with objects that they have released; they have not approached other spacecraft
 - Most RPO activities have few public details and a few remain shrouded in secrecy
 - None of the programs have strong evidence to link them to co-orbital ASAT testing or deployment
- Uncoordinated close approaches: potential for (inadvertent) escalation
 - Not as easy to make hard and fast requirements about
 - Different risk assessments by different actors in space
 - Very few hard "rules" about what is and isn't allowed



ASAT Tests by Year (2025)

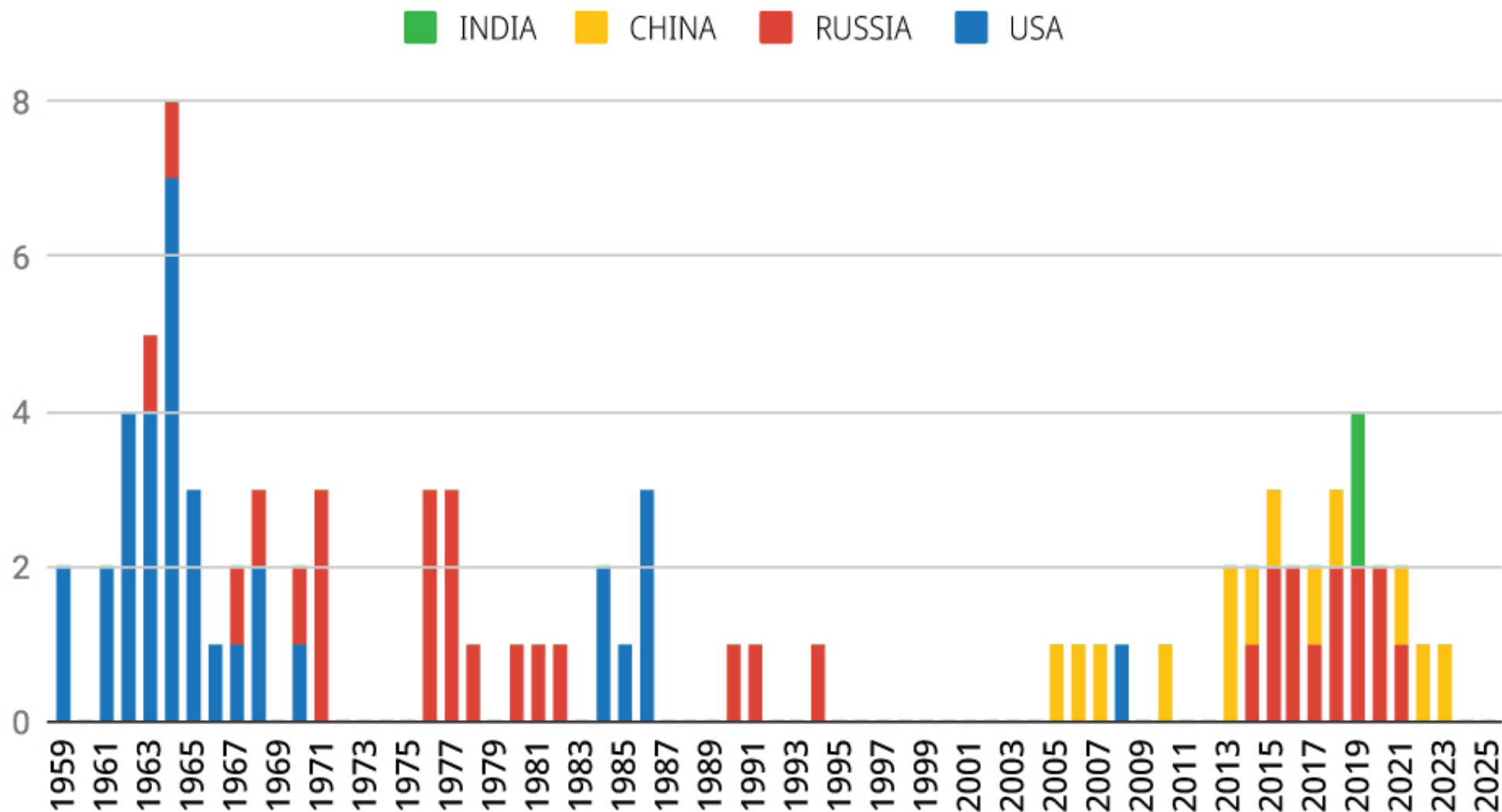




TABLE 5-1 – ORBITAL DEBRIS CREATED BY ASAT TESTS IN SPACE

DATE	COUNTRY	ASAT SYSTEM	TARGET	INTERCEPT ALTITUDE	TRACKED DEBRIS	DEBRIS STILL ON ORBIT	TOTAL DEBRIS LIFESPAN
Oct. 20, 1968	Russia	IS	Cosmos 248		252	73	50+ years
Oct. 23, 1970	Russia	IS	Cosmos 373		147	33	50+ years
Feb. 25, 1971	Russia	IS	Cosmos 394		117	44	50+ years
Dec. 3, 1971	Russia	IS	Cosmos 459		28	0	3.3 years
Dec. 17, 1976	Russia	IS	Cosmos 880		127	56	45+ years
May 19, 1978	Russia	IS-M	Cosmos 970		71	64	40+ years
Apr. 18, 1980	Russia	IS-M	Cosmos 1171		47	5	40+ years
Jun. 18, 1982	Russia	IS-M	Cosmos 1375		63	59	40+ years
Sept. 13, 1985	US	ASM-135	Solwind	530 km	285	0	18+ years
Sept. 5, 1986	US	Delta 180 PAS	Delta 2 R/B		13	0	< 1 year
Dec. 26, 1994	Russia	Naryad-V?	Unknown		26	23	30+ years
Jan. 11, 2007	China	SC-19	FengYun 1C	880 km	3533	2535	15+ years
Feb. 20, 2008	US	SM-3	USA 193	220 km	175	0	1+ year
Mar. 27, 2019	India	PDV-MK II	Microsat-R	300 km	130	0	3+ years
Aug.-Dec. 2019	Russia	Cosmos 2535	Cosmos 2536		30	16	<1 year
Nov. 15, 2021	Russia	Nudol	Cosmos 1408	470 km	1807	12	3+ years
Total					6851	2920	

REGISTRATION NOW OPEN!

7th SUMMIT for SPACE SUSTAINABILITY

OCTOBER 22-23, 2025

Centre de Conférences Pierre Mendès France
PARIS | FRANCE

More information at SWFsummit.org

#SWFsummit25

CO-HOSTED BY



SECURE
WORLD
FOUNDATION



GOVERNEMENT

*Liberté
Égalité
Fraternité*

Questions?

Thanks.

Victoria Samson,
vsamson@swfound.org



SECURE
WORLD
FOUNDATION